Updated Project Proposal - Atharv Mehendale 🚀 , Alex Seltzer 🪐 , Emmett Grossman 🐈 📦

In our project, we will simulate the collision between two elementary solar systems to learn more about the collisions and interactions between cosmic phenomena. The purpose of our project is to determine the difference, if any, in the arrangement of the final solar system depending on the angle and orientation of the solar systems colliding. Our hypothesis is that the change in the angle and orientation should have a negligible impact on the final solar system. This project was inspired by Mr. Conehat appearing in the group members' dreams and instructing them to do it. We do not want to upset Mr. Conehat.

We will write the simulation in Python, through VS Code, and visualise it using VPython. We will design and review an algorithm for simulation and the mathematics describing it in a paper procedure, then implement it in code.

In our simulation, we plan to include two solar systems composed of a star and no more than eight total rocky planets and gas giants. These will have different sizes, masses, densities, and orbital velocities based on our Sun (star), Earth (rocky planet), and Jupiter (gas giant). If time permits, we will model other planets from our solar system to add to the simulation. Initial distances and momenta will be arbitrarily chosen inspired by our own solar system. We will model the collisions between stars and planets (and with themselves) by treating them like point masses and, depending on the type of bodies involved, using roche limits. To test our hypothesis, we will vary the angle between the ecliptic planes of the solar systems in our simulation to test different orientations.

Timeline and Work Distribution

*All members will contribute equally to all parts of this project

Nov 1st - Paper Procedure completed
Nov 10th - Base Code Completed
Nov 11th - Feedback from TAs in Course Center
Nov 17nd - Final Code Completed
Nov 18rd - Final Code Checked in Course Center
Nov 22th - Print Screenshots of Simulation Images
Nov 23th - Final Touches
Nov 27th - Submit Project